

Md Saiful Islam

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Summary

HPC systems researcher with six years of prior industry software engineering experience. Builds tools that make scientific workflows portable across HPC sites and batch schedulers (Slurm, HTCondor, UGE), using declarative workflow specifications and a portable data layer on production systems including NSF ACCESS resources. Former Senior Software Engineer who led a development team.

Education

University of Notre Dame

2023 – Present

Ph.D. in Computer Science and Engineering

Notre Dame, IN

- Advisor: Douglas Thain
- Research: Portable and reproducible scientific workflows on heterogeneous HPC systems
- GPA: 3.92/4.00
- Selected coursework: Large Language Models: Basics and Practice, Graduate Operating Systems, Advanced and Secure Networks, Neural Networks, Compilers and Language Design

University of Dhaka

2013 – 2017

B.S. in Computer Science and Engineering

Dhaka, Bangladesh

Research Experience

Graduate Research Assistant

Aug 2023 – Present

Cooperative Computing Lab, University of Notre Dame

Notre Dame, IN

- **Floability** 📄: Designed and built a system that packages a workflow's code, software environment, data, and compute needs into a declarative "backpack" and deploys it across HPC sites using TaskVine. Released v0.3 with full documentation and example backpacks.
- **Portable data layer**: Designed declarative data dependencies that resolve and cache datasets across sources such as Pelican, XRootD, S3, and HTTP. Ran the same workflows unchanged across three HPC systems on 7.5 to 108 GB datasets (eScience 2026).
- **HPC site profiles**: Built hpc-site-preflight, which constructs evidence-backed site profiles from login-node probes, documentation, and pilot jobs, with LLM agents handling documentation discovery. Reached 96–99% precision across three HPC sites and catches site incompatibilities before job submission (WORKS 2026).
- **Production HPC**: Ran and evaluated workflows on Purdue Anvil, TACC Stampede3, Notre Dame CRC, and OSDF through an NSF ACCESS allocation.
- **NASA drone simulation (SADE, 2023–2024)**: Built HPC infrastructure for large-scale drone simulation in a five-university NASA project. It runs Gazebo and PX4 across cluster nodes under UGE and HTCondor. Reserved ports and resolved resource conflicts to run concurrent simulations per node, since HPC policies block isolated network namespaces. Scaled to 100 drones across 25 nodes.
- **Shepherd** 📄: Built an open-source workflow manager that runs persistent services as tasks. It infers service state from logs to coordinate startup and graceful shutdown within distributed workflows (SC24 Workshops).
- **Mentoring**: Mentored an undergraduate researcher.

Peer-Reviewed Publications

- **Md Saiful Islam** and Douglas Thain, “The Other Half of Workflow Portability: Constructing Evidence-Backed HPC Site Profiles with Agentic Discovery,” *21st Workshop on Workflows in Support of Large-Scale Science (WORKS)*, *SC26 Workshops*, 2026.
- **Md Saiful Islam**, Reyer Band, Jin Zhou, Tanu Malik, Kevin Lannon, and Douglas Thain, “Declare, Resolve, Reuse: A Portable Data Layer for Notebook Workflows Across HPC Systems,” *IEEE International Conference on eScience (eScience)*, 2026.
- **Md Saiful Islam**, Talha Azaz, Raza Ahmad, A S M Shahadat Hossain, Furqan Baig, Shaowen Wang, Kevin Lannon, Tanu Malik, and Douglas Thain, “Portable By Design: Deploying Notebook-based Scientific Workflows across HPC Clusters,” *IEEE International Conference on eScience (eScience)*, *Poster Paper*, 2026.
- Talha Azaz, Raza Ahmad, **Md Saiful Islam**, Douglas Thain, and Tanu Malik, “Efficiently Reproducing Distributed Workflows in Notebook-based Systems,” *2026 IEEE 26th International Symposium on Cluster, Cloud and Internet Computing (CCGrid)*, pp. 74–83, 2026.
- **Md Saiful Islam**, Talha Azaz, Raza Ahmad, A S M Shahadat Hossain, Furqan Baig, Shaowen Wang, Kevin Lannon, Tanu Malik, and Douglas Thain, “Backpacks for Notebooks: Enabling Containerized Notebook Workflows in Distributed Environments,” *IEEE International Conference on eScience (eScience)*, pp. 169–177, 2025.
- **Md Saiful Islam** and Douglas Thain, “Shepherd: Seamless Integration of Service Workflows into Task-Based Workflows through Log Monitoring,” *19th Workshop on Workflows in Support of Large-Scale Science (WORKS)*, *SC24 Workshops*, pp. 2080–2087, 2024.

Talks, Tutorials, and Posters

Tutorials

- “Portable by Design: Deploying Notebook-based Scientific Workflows across HPC Clusters,” three-hour tutorial, *IEEE eScience 2026*. Proposed and planned the tutorial, built the tutorial website and hands-on cloud environment, and presented remotely.

Talks

- “The Other Half of Workflow Portability,” *WORKS 2026 at SC26*, November 2026 (upcoming).
- “Floability Backpacks: Packaging Workflows for Portable HPC Deployment,” *APeX Summit 2026*, September 2026.
- “Backpacks for Notebooks,” *IEEE eScience 2025*.
- “Shepherd: Seamless Integration of Service Workflows into Task-Based Workflows,” *WORKS 2024 at SC24*.

Posters

- “Portable Execution of Data-Intensive Notebook Workflows Across Heterogeneous HPC Systems,” *GCASR 2026*.
- “Floability: Enabling Portable Scientific Workflows Across HPC Facilities,” *HPDC 2025*. Also presented at *GCASR 2025*.
- “Toward High-Performance Simulation of 1000 Drones,” *GCASR 2024*.

Industry Experience

Therap (BD) Ltd.

Senior Software Engineer

Promoted three times: Associate Software Engineer → Software Engineer → Software Engineer II → Senior Software Engineer

Apr 2017 – Jul 2023

Dhaka, Bangladesh

- **Leadership:** Led a team of 5 engineers and co-led company-wide project teams. Interviewed and hired developers.
- **Requirements and design:** Worked with product, marketing, and training teams to gather requirements,

design new products, and update existing features.

- **Code review:** Served as a primary reviewer of source code and test cases for every release, catching design, correctness, and security issues before production.
- **Interoperability and compliance:** Implemented health interoperability standards and modernized the company's health information exchange platform with HIPAA-compliant data services.
- **Smart devices and ML:** Early member of Therap Connect. Integrated smart devices for fall detection and intelligent alerts. Briefly contributed to launching the company's machine learning group.
- **Performance:** Improved performance and stability of large batch applications on a production healthcare platform.
- **Stack:** Java, Spring, Hibernate

Technical Skills

Languages: Python, Java, C, C++ , Bash

HPC and Schedulers: Slurm, HTCondor, UGE/SGE, NSF ACCESS resources

Workflow Systems: TaskVine, Makeflow, Parsl, Dask

Containers and Environments: Apptainer, Podman, Charliecloud, Docker, Conda

Data Access: Pelican, XRootD, S3

AI and Agents: LLM agents, tool calling, LangChain, OpenAI and Gemini APIs, Ollama, LM Studio, PyTorch

Web and Backend: Spring, Hibernate, Django, Flask, React

Cloud and Tools: AWS EC2, Linux, Git

Selected Course Projects

- **Large Language Models:** Co-designed AETHER, an evaluation of hallucinations in LLM-generated Python and Java code across five LLMs, using execution, unit tests, and static analysis. Feedback-oriented prompting cut hallucination rates by about half.
- **Neural Networks:** Built a U-Net speech denoising autoencoder in PyTorch, trained on 16,910 noisy LibriSpeech segments. [↗](#)
- **Advanced and Secure Networks:** Implemented an NDN traceroute client and producer in C++ following RFC 9507. [↗](#)

Service and Leadership

- **Student Volunteer,** ACM HPDC 2025, Notre Dame, IN. Coordinated sessions and hosted invited guests.
- **Treasurer,** Bangladeshi Students Association of Notre Dame, 2024–2025. Managed budgets and logistics for 10+ cultural events a year.
- **Co-founder,** E-Boipotro, 2016–Present. Built an open-source EPUB library of public-domain Bengali books, including the complete works of Rabindranath Tagore. Organized a crowdsourced digitization effort.

References

Dr. Douglas Thain

Professor

Department of Computer Science and Engineering

University of Notre Dame, IN, USA

Email: dthain@nd.edu

Relation: Ph.D. Advisor